

Most essential, however, is that since the trade schedule is expressed in terms of a continuous exponential function the analytical gradient and Hessian can be easily computed. This dramatically increases the computational efficiency of a non-linear optimization algorithm. Finally, it incorporates the investor's exact risk aversion parameter.

A limitation of the exponential trade schedule, however, is that it does not allow as much freedom to take advantage of natural hedging and diversification as the exact NLP and QP approaches described above. The lower and upper bounds are included on the trading schedule parameter to ensure the order is traded within a user specified rate.

Residual Schedule Exponential

The residual schedule exponential is a technique that parameterizes the residual schedule in terms of an exponential decay function. It is a non-linear optimization routine and uses the square root function for the risk term.

The residual number of shares in each period is determined by the following:

$$r_{ij} = X_i \cdot e^{-j\omega_i} \quad (9.10)$$

This formulation is a decreasing function so it will always adhere to our decreasing portfolio constraint. But since it is always positive, i.e., $r_{ij} > 0$, we need to incorporate some terminal value to force the order to complete (within some tolerance).

The optimization formulation for the trade schedule exponential is:

$$\text{Min}_x \sum_{i=1}^m \sum_{t=1}^{n+1} \frac{b_1 \cdot I_i^* \cdot (r_{it} - r_{it+1})^2}{X_i \cdot v_{it}} + \lambda \cdot \sqrt{\sum_{k=1}^{n+1} r'_k C r_k} \quad (9.11)$$

Subject to:

$$\begin{array}{ll} \overline{r_{ij} = X_i \cdot e^{-j\omega_i}} & \text{for all } i, j \\ r_{i1} = X_i & \text{for all } i \\ r_{in+1} \leq 100 & \text{for all } i \\ x_{ij} = r_{ij} - r_{ij+1} & \text{for all } i, j \\ \underline{LB_i} \leq \omega_i \leq \overline{UB_i} & \text{for all } i \end{array}$$

Expressing the trade schedule as a parametric exponential formulation provides many benefits.

First, there is only one parameter to estimate for each stock regardless of the number of specified trading periods and trading days. For example,